FACULTY OF COMPUTING, ELECTRONICS AND SCIENCE ASSESSMENT

PRINCIPLES OF COMPUTING

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Entity Relationship Diagram (ERD) - Text Representation:

An ERD visually represents the entities, relationships between them, and attributes. Here's a textual representation of the ERD for Peter Darwin Estate Agency based on your SQL script:

Entities:

- Branch (BranchID [PK], BranchName, BranchLocation, ManagerName)

Connections | 1.IS4s671.sql | BRANCH

Columns | Data | Model | Constraints | Grants | Statistics | Triggers | Flashback | Dependencies | Details | Partitions | Indexes | SQL

Actions...

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	BRANCHID	NUMBER(38,0)	No	(null)	1	(null)
2	BRANCHNAME	VARCHAR2(50 BYTE)	Yes	(null)	2	(null)
3	BRANCHLOCATION	VARCHAR2(100 BYTE)	Yes	(null)	3	(null)
4	MANAGERNAME	VARCHAR2(50 BYTE)	Yes	(null)	4	(null)

- Staff (StaffID [PK], BranchID [FK], StaffName, Position)

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	STAFFID	NUMBER(38,0)	No	(null)	1	(null)
2	BRANCHID	NUMBER(38,0)	Yes	(null)	2	(null)
3	STAFFNAME	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	POSITION	VARCHAR2(50 BYTE)	Yes	(null)	4	(null)

- Client (ClientID [PK], ClientName, MaxLeaseLength, PetsAllowed, ChildrenAllowed)

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	CLIENTID	NUMBER(38,0)	No	(null)	1	(null)
2	CLIENTNAME	VARCHAR2(50 BYTE)	Yes	(null)	2	(null)
3	MAXLEASELENGTH	NUMBER(38,0)	Yes	(null)	3	(null)
4	PETSALLOWED	CHAR(1 BYTE)	Yes	(null)	4	(null)
5	CHILDRENALLOWED	CHAR(1 BYTE)	Yes	(null)	5	(null)

- RentalProperty (PropertyCode [PK], BranchID [FK], Area, Address, Bedrooms, Bathrooms, ReceptionRooms, HasGarage, PropertyType, MonthlyRental)

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	PROPERTYCODE	NUMBER(38,0)	No	(null)	1	(null)
2	BRANCHID	NUMBER(38,0)	Yes	(null)	2	(null)
3	AREA	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	ADDRESS	VARCHAR2(100 BYTE)	Yes	(null)	4	(null)
5	BEDROOMS	NUMBER(38,0)	Yes	(null)	5	(null)
6	BATHROOMS	NUMBER(38,0)	Yes	(null)	6	(null)
7	RECEPTIONROOMS	NUMBER(38,0)	Yes	(null)	7	(null)
8	HASGARAGE	CHAR(1 BYTE)	Yes	(null)	8	(null)
9	PROPERTYTYPE	VARCHAR2(20 BYTE)	Yes	(null)	9	(null)
10	MONTHLYRENTAL	NUMBER(10,2)	Yes	(null)	10	(null)

- RentalCustomer (RentalCustomerID [PK], PropertyCode [FK], ClientID [FK])

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	RENTALCUSTOMERID	NUMBER(38,0)	No	(null)	1	(null)
2	PROPERTYCODE	NUMBER(38,0)	Yes	(null)	2	(null)
3	CLIENTID	NUMBER(38,0)	Yes	(null)	3	(null)

- PropertyForSale (PropertyCode [PK], BranchID [FK], Area, Address, Bedrooms, Bathrooms, ReceptionRooms, HasGarage, PropertyType, IsFreehold, Price, CurrentState, DateOnMarket)

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	PROPERTYCODE	NUMBER(38,0)	No	(null)	1	(null)
2	BRANCHID	NUMBER(38,0)	Yes	(null)	2	(null)
3	AREA	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	ADDRESS	VARCHAR2(100 BYTE)	Yes	(null)	4	(null)
5	BEDROOMS	NUMBER(38,0)	Yes	(null)	5	(null)
6	BATHROOMS	NUMBER(38,0)	Yes	(null)	6	(null)
7	RECEPTIONROOMS	NUMBER(38,0)	Yes	(null)	7	(null)
8	HASGARAGE	CHAR(1 BYTE)	Yes	(null)	8	(null)
9	PROPERTYTYPE	VARCHAR2(20 BYTE)	Yes	(null)	9	(null)
10	ISFREEHOLD	CHAR(1 BYTE)	Yes	(null)	10	(null)
11	PRICE	NUMBER(10,2)	Yes	(null)	11	(null)
12	CURRENTSTATE	VARCHAR2(10 BYTE)	Yes	(null)	12	(null)
13	DATEONMARKET	DATE	Yes	(null)	13	(null)

- PotentialBuyer (BuyerID [PK], Name, Address)

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	BUYERID	NUMBER(38,0)	No	(null)	1	(null)
2	NAME	VARCHAR2(50 BYTE)	Yes	(null)	2	(null)
3	ADDRESS	VARCHAR2(100 BYTE)	Yes	(null)	3	(null)

- PropertyRequirements (RequirementID [PK], BuyerID [FK], Area, MinBedrooms, MinBathrooms, MinReceptionRooms, GarageRequired, MaxPrice)

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	REQUIREMENTID	NUMBER(38,0)	No	(null)	1	(null)
2	BUYERID	NUMBER(38,0)	Yes	(null)	2	(null)
3	AREA	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	MINBEDROOMS	NUMBER(38,0)	Yes	(null)	4	(null)
5	MINBATHROOMS	NUMBER(38,0)	Yes	(null)	5	(null)
6	MINRECEPTIONROOMS	NUMBER(38,0)	Yes	(null)	6	(null)
7	GARAGEREQUIRED	CHAR(1 BYTE)	Yes	(null)	7	(null)
8	MAXPRICE	NUMBER(10,2)	Yes	(null)	8	(null)

- ViewingAppointment (AppointmentID [PK], PropertyCode [FK], StaffID [FK])

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	APPOINTMENTID	NUMBER(38,0)	No	(null)	1	(null)
2	PROPERTYCODE	NUMBER(38,0)	Yes	(null)	2	(null)
3	STAFFID	NUMBER(38,0)	Yes	(null)	3	(null)

- Offer (OfferID [PK], PropertyCode [FK], BuyerID [FK], Status)

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	OFFERID	NUMBER(38,0)	No	(null)	1	(null)
2	PROPERTYCODE	NUMBER(38,0)	Yes	(null)	2	(null)
3	BUYERID	NUMBER(38,0)	Yes	(null)	3	(null)
4	STATUS	VARCHAR2(10 BYTE)	Yes	(null)	4	(null)

Relationships:

- Branch - Staff: One-to-Many
- Branch - RentalProperty: One-to-Many
- RentalProperty - RentalCustomer: One-to-Many
- Branch - PropertyForSale: One-to-Many
- PropertyForSale - PotentialBuyer: One-to-Many
- PotentialBuyer - PropertyRequirements: One-to-Many
- PropertyForSale - ViewingAppointment: One-to-Many
- PropertyForSale - Offer: One-to-Many

Primary and Foreign Keys:

- **Branch:**

Primary Key: BranchID

- **Staff:**

Primary Key: StaffID

Foreign Key: BranchID (references Branch(BranchID))

- **Client:**

Primary Key: ClientID

- **RentalProperty:**

Primary Key: PropertyCode

Foreign Key: BranchID (references Branch(BranchID))

- **RentalCustomer:**

Primary Key: RentalCustomerID

Foreign Keys: PropertyCode (references RentalProperty(PropertyCode)), ClientID (references Client(ClientID))

- **PropertyForSale:**

Primary Key: PropertyCode

Foreign Key: BranchID (references Branch(BranchID))

- **PotentialBuyer:**

Primary Key: BuyerID

- **PropertyRequirements:**

Primary Key: RequirementID

Foreign Key: BuyerID (references PotentialBuyer(BuyerID))

- **ViewingAppointment:**

Primary Key: AppointmentID

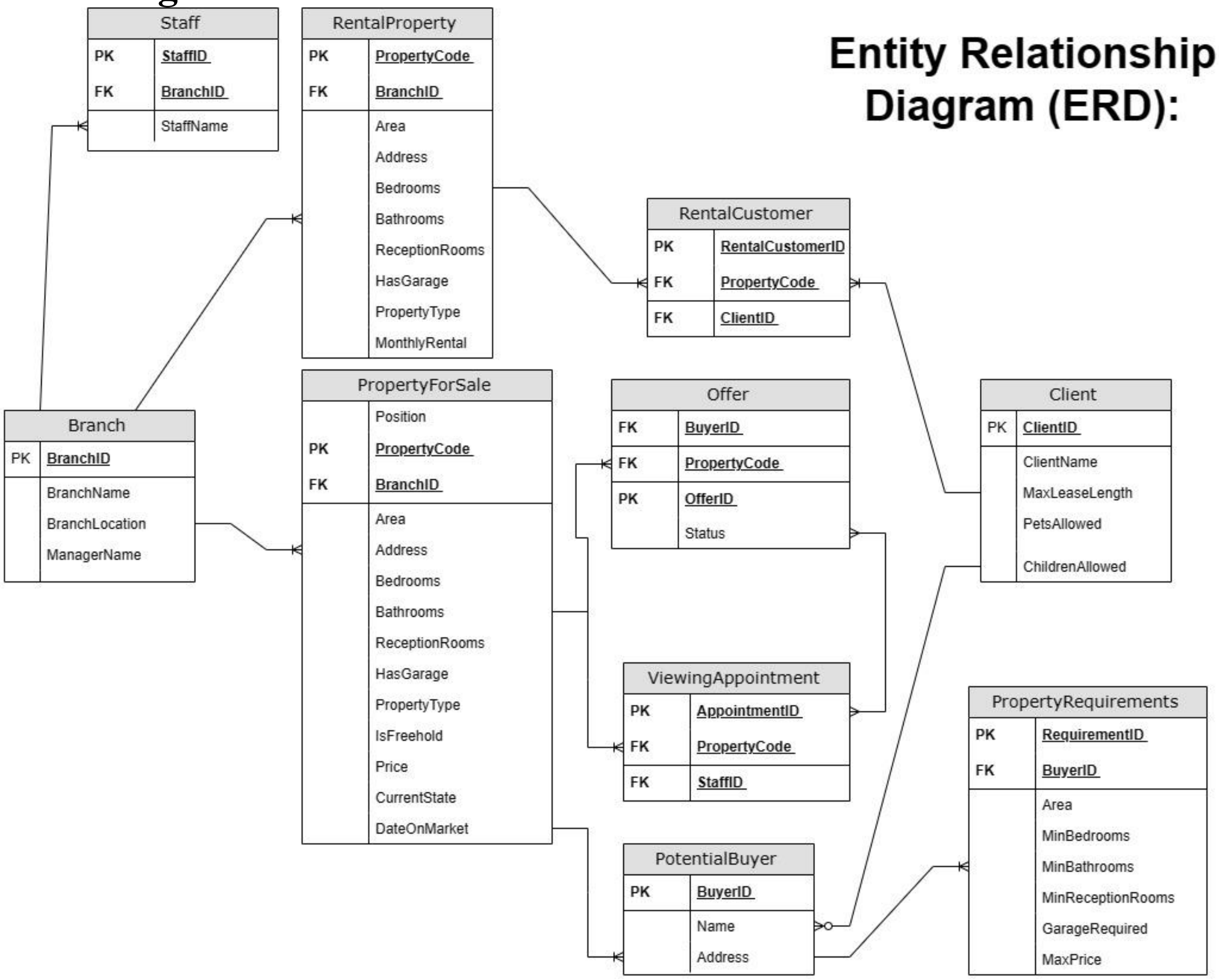
Foreign Keys: PropertyCode (references PropertyForSale(PropertyCode)), StaffID (references Staff(StaffID))

- **Offer:**

Primary Key: OfferID

Foreign Keys: PropertyCode (references PropertyForSale(PropertyCode)), BuyerID (references PotentialBuyer(BuyerID))

ER Diagram:



Assumptions:

1. **Assumption:** A branch can have multiple staff members, but each staff member belongs to only one branch.
    - **Why:** This assumption is based on the common organizational structure where staff members are assigned to specific branches.
2. **Assumption:** Rental and sale properties have different sets of attributes.
    - **Why:** The scenario did not explicitly mention shared attributes between rental and sale properties, so it's assumed they have distinct sets of characteristics.
3. **Assumption:** Clients can rent multiple properties, and each rental property is associated with one client.
    - **Why:** This assumption aligns with the scenario's statement that details are recorded about each client and on each property, they wish to rent.

4. **Assumption:** Each viewing appointment is assigned to one staff member, and a staff member can have multiple viewing appointments.
    - **Why:** This assumption is made based on the scenario's mention that a member of staff is assigned to accompany prospective buyers to each viewing appointment.
5. **Assumption:** An offer is associated with one property and one buyer.
    - **Why:** This assumption is based on the scenario's description that when a buyer puts in an offer for a property, the seller is informed.

Conclusion:

In conclusion, the completion of Milestone 1 for Coursework 1 in the Principles of Computing module has resulted in a comprehensive understanding and representation of the database design for Peter Darwin Estate Agency. The Entity Relationship Diagram (ERD) successfully captures the relationships between entities, their attributes, and the intricacies outlined in the provided scenario from Appendix A.

The inclusion of Primary and Foreign keys in the ERD ensures the establishment of accurate relationships between entities, contributing to the overall integrity of the database design. The keys serve as vital components in maintaining referential integrity and fostering efficient data management within the system.

Assumptions made during the development process are well-documented, providing clarity on the decisions taken during the design phase. These assumptions, ranging from organizational structures to property-specific attributes, were essential in addressing ambiguities within the scenario and enhancing the completeness of the ERD.